

Lab Session Runbook

SE498 Team 3 “Audio Rate” · one participant, one visit, about 65 minutes · state as of 2026-09-16 · every step traces to an accepted decision; [TBD] marks what the team still fills in

For future Claude

The experimenter-facing procedure for a collection session, assembled from the protocol decisions of 2026-09-16 (Decision - Protocol one session per participant age-predicted HRmax · Decision - Recovery window recorded from session one · Decision - Labels by fitted lag continuous exertion trajectory · Decision - Breathing-rate reference H10 first belt conditional · Decision - IMU logged by the capture app · Decision - Handset roster three iPhones) and the capture rules in Audio-Derived Exercise Exertion Estimation §1–4. Companion documents: Consent Form - Audio Rate (what the participant signs), Data Management Plan (where the files go), Pipeline Input to Verdict (what happens to the data). PDF beside this note is footer-free so it can be printed for the lab. Update when an ADR changes; re-export with `--no-foot`.

0. Before the day

Item	Status	Owner
Ergometer: Jude’s Wahoo KICKR Core, ERG mode, bike mounted, controlling app on a device other than the strap-ingest phone (Decision - Jude’s Wahoo KICKR as the cycle ergometer)	[TBD]	Jude
Quiet room booked (Risk 5) — settled 09-22: the small apartment room where the KICKR Core lives; confirm it is free for the slot, HVAC and fans off during recording	[TBD]	[TBD]
Polar H10 in hand; nRF Connect shows the RR flag; Polar SDK streams ECG + ACC	[TBD]	Josh
Three iPhones listed (model, iOS version), each passed the qualification test in the last 30 days	[TBD]	[TBD]
Capture app: <code>.measurement</code> mode, voice processing off, 16 kHz WAV, IMU logging, H10 ingestion, event buttons, file-injection parity test green	[TBD]	Josh
Wired headset boom mic + foam windscreen; calibration-tone source; tape for the fixed geometry	[TBD]	[TBD]
Printed consent form ×1, health screen ×1, Borg 6–20 card, differentiated-rating card, session sheet	[TBD]	[TBD]
Participant code assigned in the key sheet (kept outside the vault)	[TBD]	Josh
Water bottle for the participant; thermometer or the room’s reading	[TBD]	[TBD]

1. Arrival and consent (10 min, not recorded)

1. Welcome; read the script in Participant Consent Script and Form aloud; answer questions.
2. Health screen first (§8 of the form). Any “yes” → thank them, stop, no data.
3. Consent initials and signatures. Form goes in the consent folder, never near the data.

- Record on the session sheet: participant code, date, **room temperature**, water offered (yes/no), caffeine in the last 3 h, hours of sleep, training background if the optional line was initialled (Decision - Protocol one session per participant age-predicted HRmax).
- Compute Tanaka HRmax = $208 - 0.7 \times \text{age}$; write it on the sheet. Never ask the participant to ride to exhaustion.

2. Setup (10 min)

- Strap:** wet the electrodes; fit the Polar H10 under the shirt, snug. Start the app's H10 stream; confirm RR intervals and ECG arrive; watch for the dry-electrode garbage to clear.
- Microphone:** wired headset boom at the fixed geometry — near the mouth corner, off-axis from the nostrils, windscreen on. Tape the boom position. Plug into iPhone A. Mount iPhone B (and C if used) at the logged distance [TBD] on the handlebar mount.
- Gain:** set the input gain to the study value [TBD] and **do not touch it again**. Log the value. A mid-session gain change invalidates the session's level comparisons.
- App readback:** confirm `.measurement` mode, voice processing off, input route, sample rate, IMU on. Any failed readback is a hard stop, not a warning.
- Seated rest begins as soon as the strap is stable — see §3.

3. Recorded session (≈ 45 min riding + rest)

Start **one continuous recording per phone** at the beginning of seated rest and stop it after the final recovery. Press the app's event button at every transition; write the clock time on the sheet as a backup.

Step	Duration	What the experimenter does	Event
Seated rest	5 min	Participant sits still, no talking. Resting HR = last 3 min.	<code>rest_start</code>
Pre-roll	1 min	Calibration tone from the reference source at the taped distance (10 s) → 10 s room tone (silence) → sync clap : one hand clap while the other hand touches the strap electrodes, visible in audio and RR.	<code>cal_tone</code> , <code>room_tone</code> , <code>clap</code>
Practice ride	3 min	Light load. Explain the Borg 6–20 scale and the three differentiated ratings (overall / legs / breathing). Take one practice rating.	<code>practice_start</code>
Warm-up	3 min	Fixed low wattage [TBD]. Standardised starting state; not a lag fix.	<code>warmup_start</code>
Stages	3–4 × 3–4 min	Wattage steps [TBD] (rising to “hard”, RPE ≈ 15 –16). In the last 30 s of every stage : Borg overall, legs, breathing. Participant points; experimenter says the numbers aloud once (they are on the recording; fine) and writes them.	<code>stage_N_start</code> , <code>rpe_N</code>
Recovery 1	3–5 min	Unloaded pedalling or seated. Keep recording — this is D1 data (Decision - Recovery window recorded from session one).	<code>recovery1_start</code>

Step	Duration	What the experimenter does	Event
Drift bout	15–20 min	Constant moderate load [TBD] W ($\approx 60\%$ of the participant's top stage). Ratings every 5 min and at the end. Extend from 15 to 20 min if the participant and the clock allow. Water available.	bout_start, rpe_bout_k
Recovery 2	3–5 min	As Recovery 1.	recovery2_start
Stop	—	Stop recordings on every phone, then the H10 stream.	session_end

Stop rules, spoken before the practice ride: the participant says “stop” · chest discomfort · dizziness · unusual breathlessness · pain · the experimenter’s judgement. End immediately, keep them seated, let them cool down, log `aborted` and the reason. Aborted sessions are kept but flagged; nothing is “rescued”.

Silence rule: no conversation during recorded steps except the rating exchange. If the participant speaks, note the time; the speech review in Data Management Plan strips it.

4. Teardown (5 min)

1. Strap off; participant offered water; thank them; remind them of the deletion deadline on the form.
2. Copy the session bundle from every phone to the encrypted drive **before leaving the room** (Data Management Plan §2). Verify file sizes are non-zero and durations match.
3. Fill the QA sheet (§5). Wipe the boom windscreen or swap it.

5. Same-day QA (30–60 min, experimenter or Josh)

A session enters modelling only when every line passes. Reject, do not rescue.

- [] Calibration tone level within ± 1 dB of the study reference on every phone; room-tone floor did not gate to silence; no spectral hole 1–4 kHz (design note §2).
- [] Sync clap located in every audio file and in the RR/ECG stream; offsets written to the bundle metadata.
- [] Event log complete; stage times agree with the sheet within 5 s.
- [] RR rejection rate below [TBD] % per stage; no cadence-locking (HR tracking pedal cadence).
- [] Lag fit runs and lands inside 0–90 s with a single clear peak (Decision - Labels by fitted lag continuous exertion trajectory; Risk 10).
- [] H10-derived breathing rate (ECG and ACC) computed for the bout and the rest period; **pilot one only:** ± 3 br/min vs a manual count at rest and at the top stage decides the belt (Decision - Breathing-rate reference H10 first belt conditional).
- [] Speech segments flagged by VAD and reviewed; strip list written.
- [] IMU file present for every phone, timestamps on the audio clock.

6. Session sheet fields (paper, then typed into the bundle's `metadata.json`)

participant code · date · experimenter · room temperature · water offered / intake · caffeine (h since) · sleep (h) · age (for Tanaka) · training background (optional) · Tanaka HRmax · resting HR · phones used (model, iOS, mount, gain) · boom distance · wattage per stage · bout wattage · ratings table · events with clock times · stop rule used (if any) · QA outcome.

Connections

- Consent Form - Audio Rate · Participant Consent Script and Form · Data Management Plan · Pipeline Input to Verdict · Equipment and Purchase List · Audio-Derived Exercise Exertion Estimation §1–4 · Ground Truth and Validation · Synthesis/Risk Register rows 5, 8, 10, 11 · Boards/Tasks

Connections — auto-woven (heal)

Domain: Synthesis/Method MOC

Structural (live):

```
LIST WHERE type = this.type AND file.name != this.file.name SORT file.name ASC
```

```
LIST WHERE contains(file.outlinks, this.file.link) SORT file.name ASC
```

Semantic: - Sensing Requirements and Signal Conditioning — ten-question answers the runbook realises - Milestones — pilot one runs this end to end